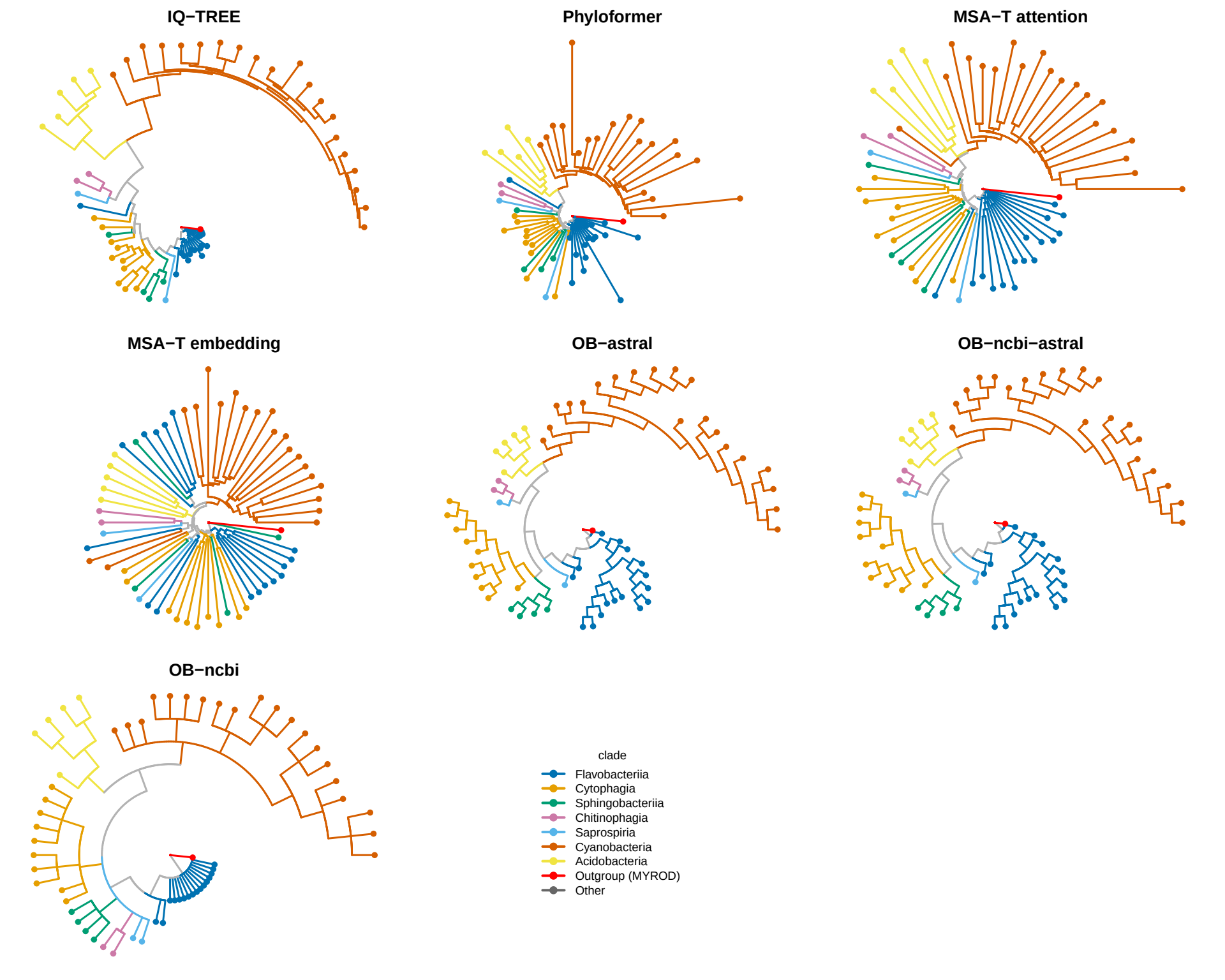
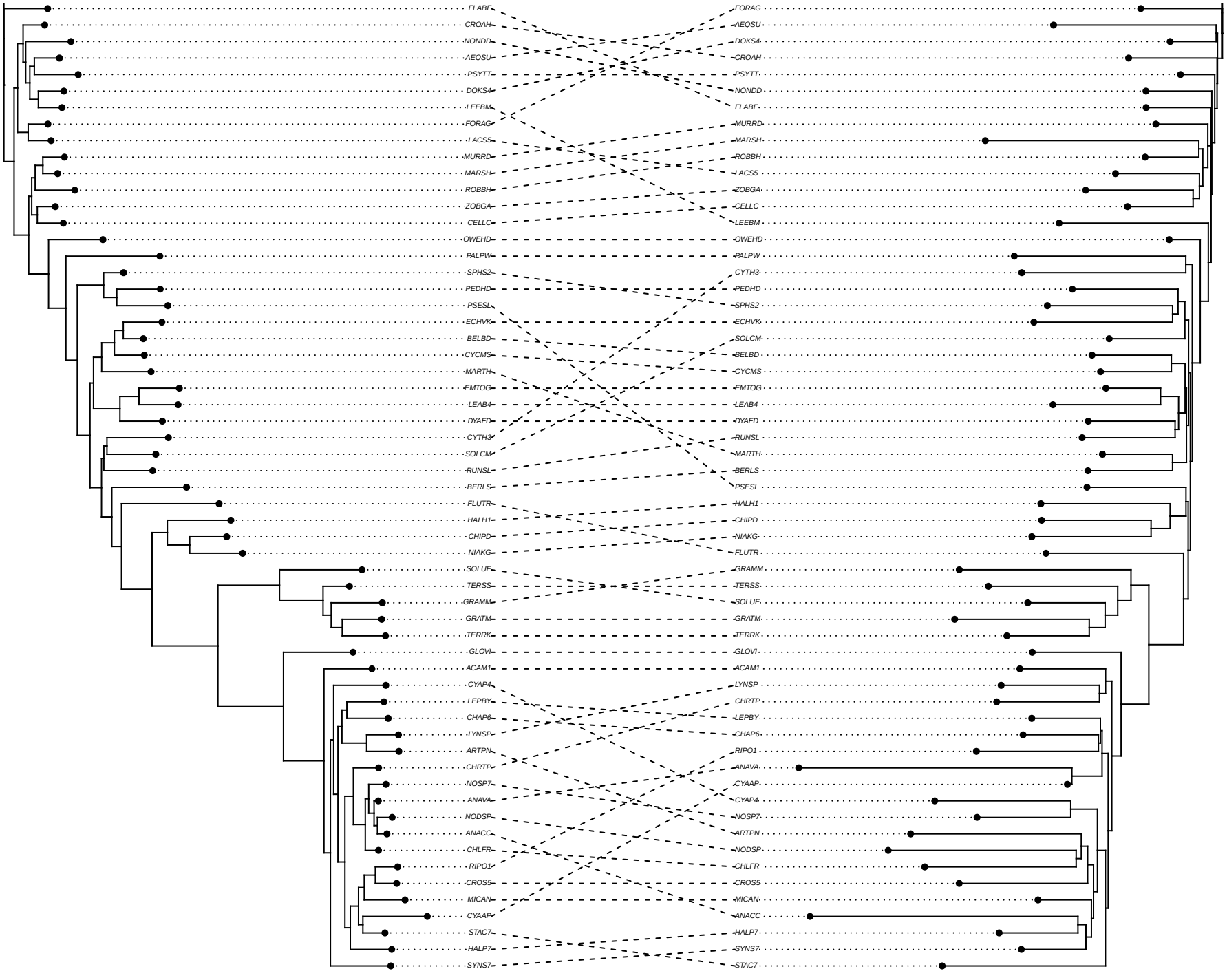
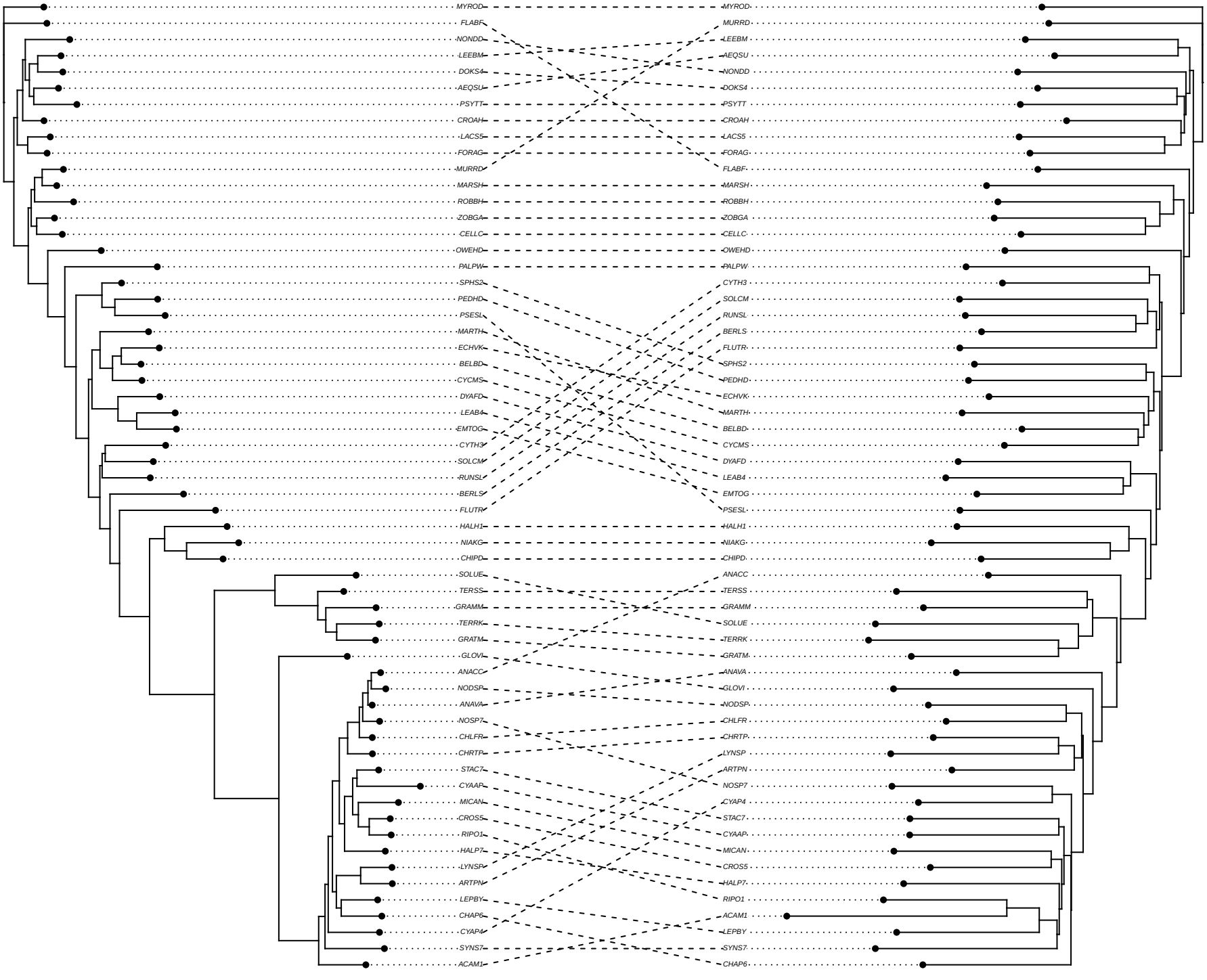
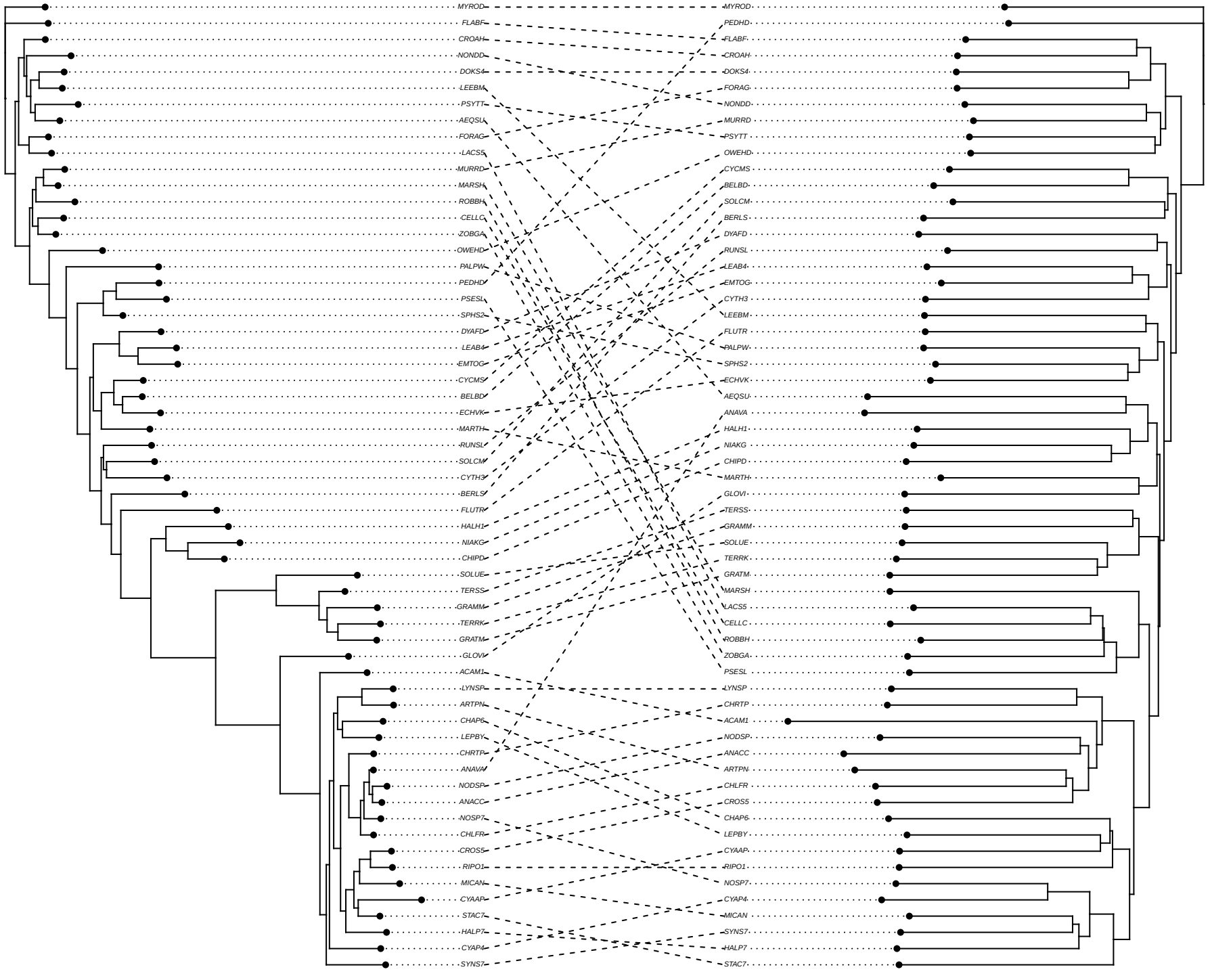

IQ−TREE
Phyloformer
MSA−T attention
MSA−T embedding
OB−astral
OB−ncbi−astral
OB−ncbi
clade
Flavobacteriia
Cytophagia
Sphingobacteriia
Chitinophagia
Saprospiria
Cyanobacteria
Acidobacteria
Outgroup (MYROD)
Other

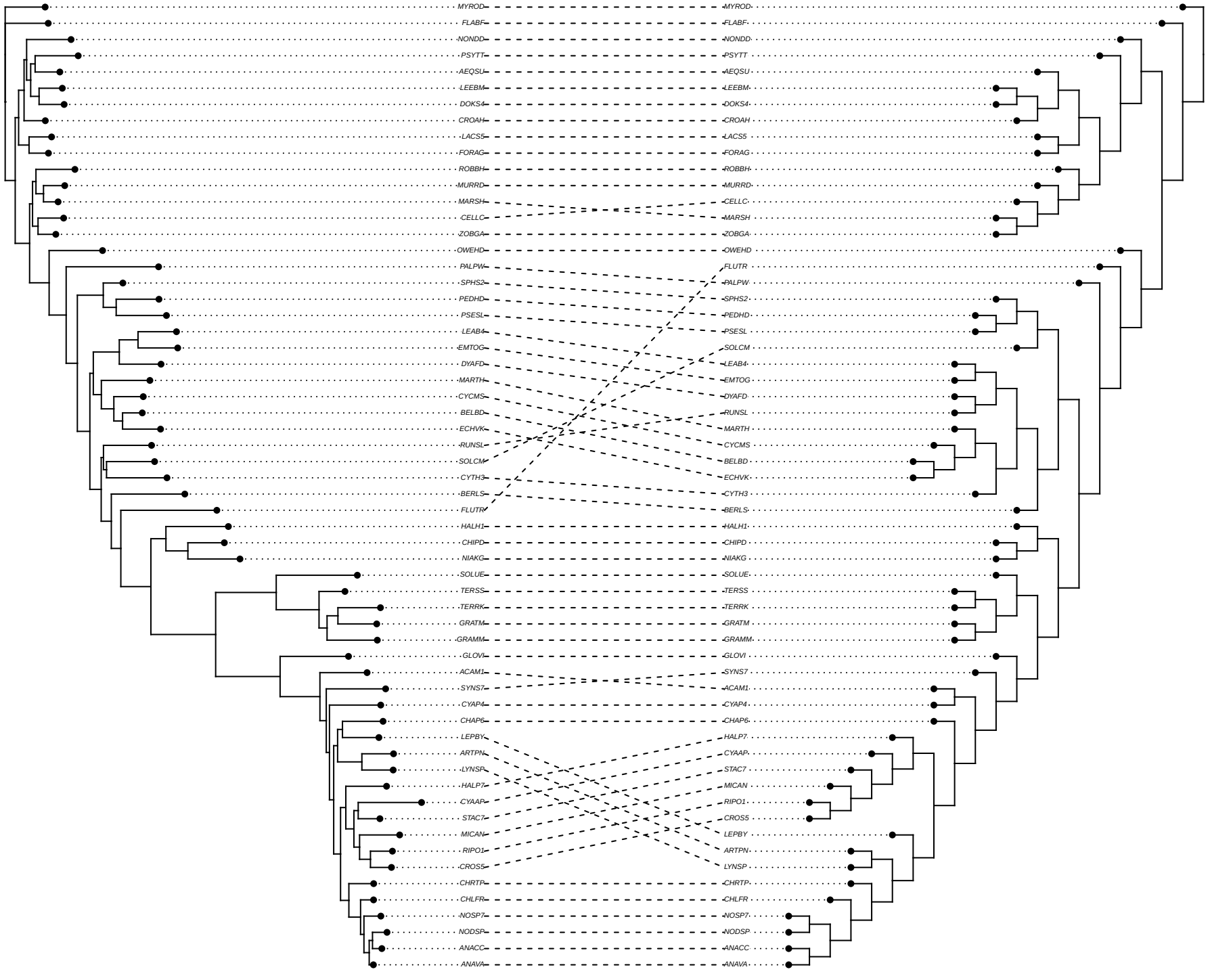
IQ-TREE vs Phylolformer



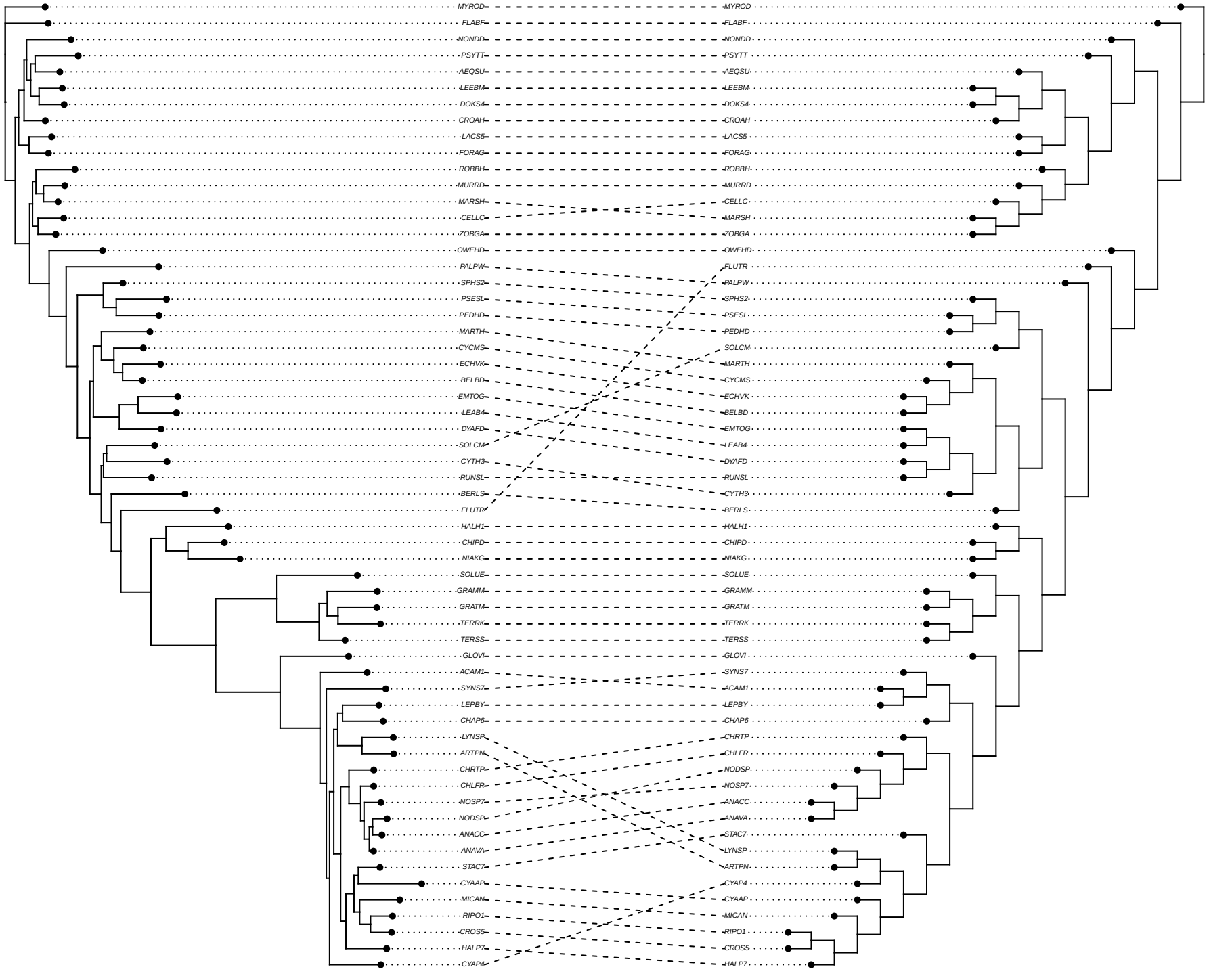

IQ−TREE vs MSA−Pattention
MYROD
FLABF
NONDD
LEEBM
DOKS4
AEQSU
PSYTT
CROAH
LACS5
FORAG
MURRD
MARSH
ROBBH
ZOBGA
CELLC
OWEHD
PALPW
SPHS2
PEDHD
PSESL
MARTH
ECHVK
BELBD
CYCMS
DYAFD
LEAB4
EMTOG
CYTH3
SOLCM
RUNSL
BERLS
FLUTR
HALH1
NIAKG
CHIPD
SOLUE
TERSS
GRAMM
TERRK
GRATM
GLOVI
ANACC
NODSP
ANAVA
NOSP7
CHLFR
CHRTP
STAC7
CYAAP
MICAN
CROS5
RIPO1
HALP7
LYNSP
ARTPN
LEPBY
CHAP6
CYAP4
SYNS7
ACAM1
MYROD
MURRD
LEEBM
AEQSU
NONDD
DOKS4
PSYTT
CROAH
LACS5
FORAG
FLABF
MARSH
ROBBH
ZOBGA
CELLC
OWEHD
PALPW
CYTH3
SOLCM
RUNSL
BERLS
FLUTR
SPHS2
PEDHD
ECHVK
MARTH
BELBD
CYCMS
DYAFD
LEAB4
EMTOG
PSESL
HALH1
NIAKG
CHIPD
ANACC
TERSS
GRAMM
SOLUE
TERRK
GRATM
ANAVA
GLOVI
NODSP
CHLFR
CHRTP
LYNSP
ARTPN
NOSP7
CYAP4
STAC7
CYAAP
MICAN
CROS5
HALP7
RIPO1
ACAM1
LEPBY
SYNS7
CHAP6


IQ−TREE vs MSA−T embedding
MYROD
FLABF
CROAH
NONDD
DOKS4
LEEBM
PSYTT
AEQSU
FORAG
LACS5
MURRD
MARSH
ROBBH
CELLC
ZOBGA
OWEHD
PALPW
PEDHD
PSESL
SPHS2
DYAFD
LEAB4
EMTOG
CYCMS
BELBD
ECHVK
MARTH
RUNSL
SOLCM
CYTH3
BERLS
FLUTR
HALH1
NIAKG
CHIPD
SOLUE
TERSS
GRAMM
TERRK
GRATM
GLOVI
ACAM1
LYNSP
ARTPN
CHAP6
LEPBY
CHRTP
ANAVA
NODSP
ANACC
NOSP7
CHLFR
CROS5
RIPO1
MICAN
CYAAP
STAC7
HALP7
CYAP4
SYNS7
MYROD
PEDHD
FLABF
CROAH
DOKS4
FORAG
NONDD
MURRD
PSYTT
OWEHD
CYCMS
BELBD
SOLCM
BERLS
DYAFD
RUNSL
LEAB4
EMTOG
CYTH3
LEEBM
FLUTR
PALPW
SPHS2
ECHVK
AEQSU
ANAVA
HALH1
NIAKG
CHIPD
MARTH
GLOVI
TERSS
GRAMM
SOLUE
TERRK
GRATM
MARSH
LACS5
CELLC
ROBBH
ZOBGA
PSESL
LYNSP
CHRTP
ACAM1
NODSP
ANACC
ARTPN
CHLFR
CROS5
CHAP6
LEPBY
CYAAP
RIPO1
NOSP7
CYAP4
MICAN
SYNS7
HALP7
STAC7

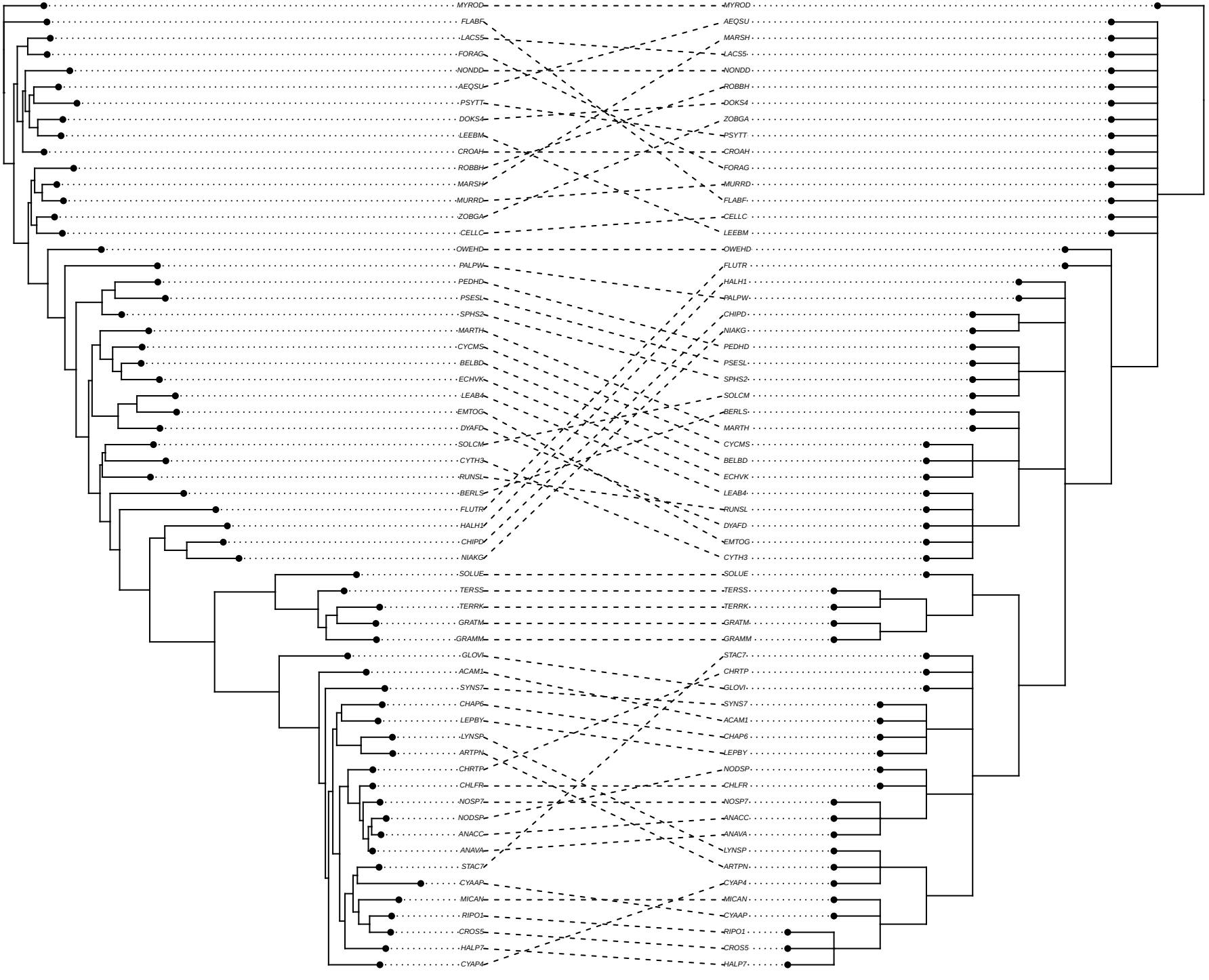
IQ−TREE vs OB−astral

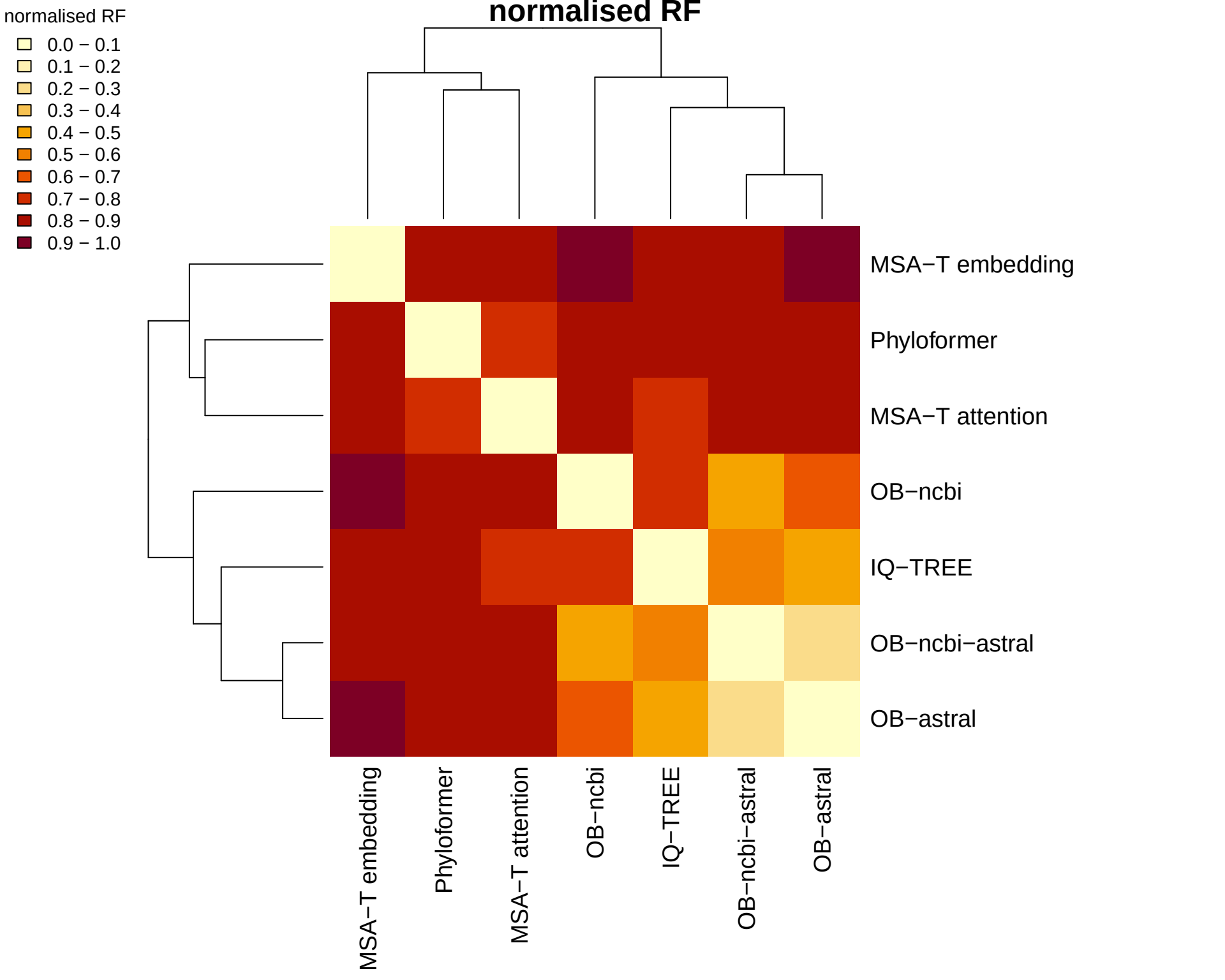


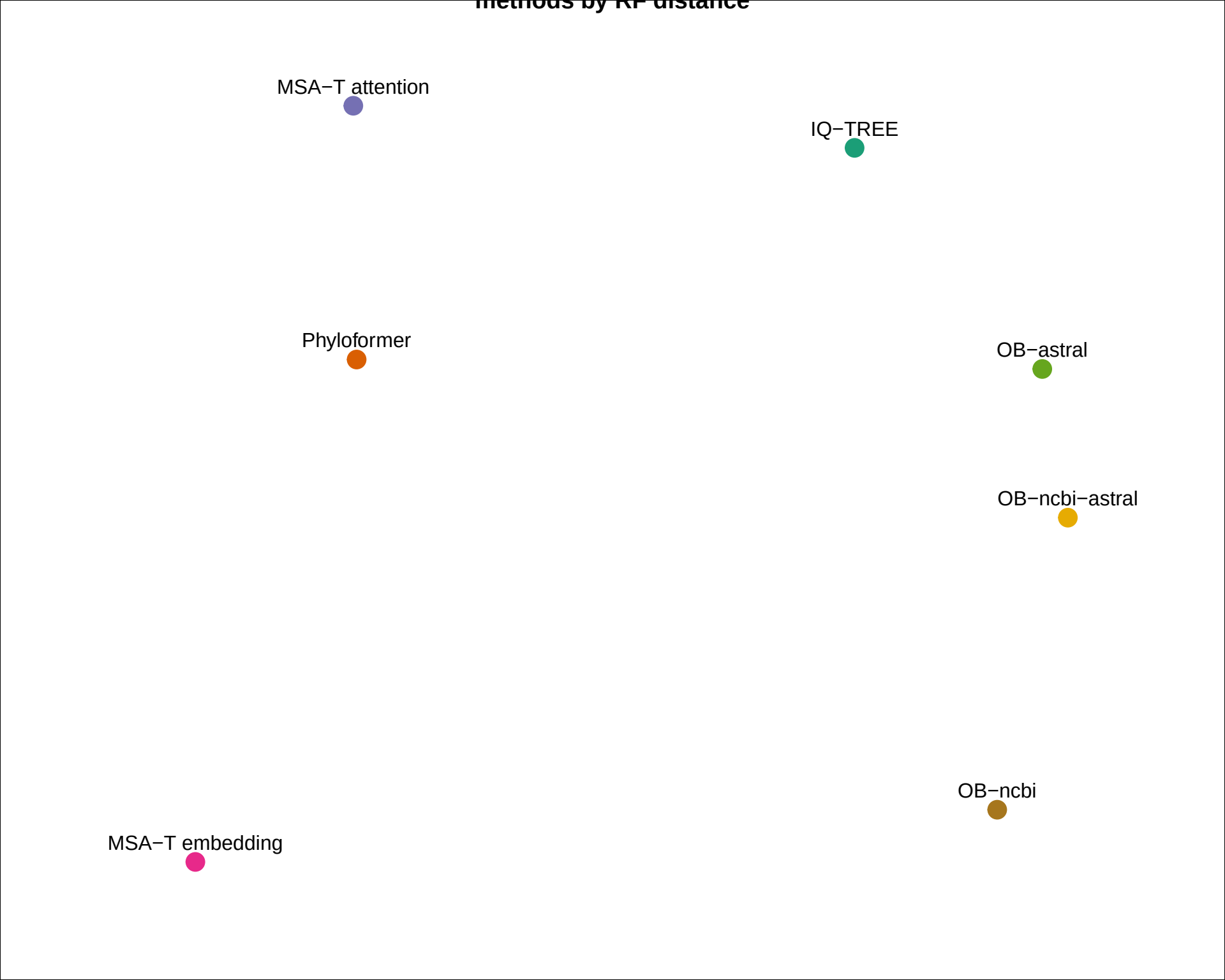
IQ-TREE vs OB-ncbi-astral

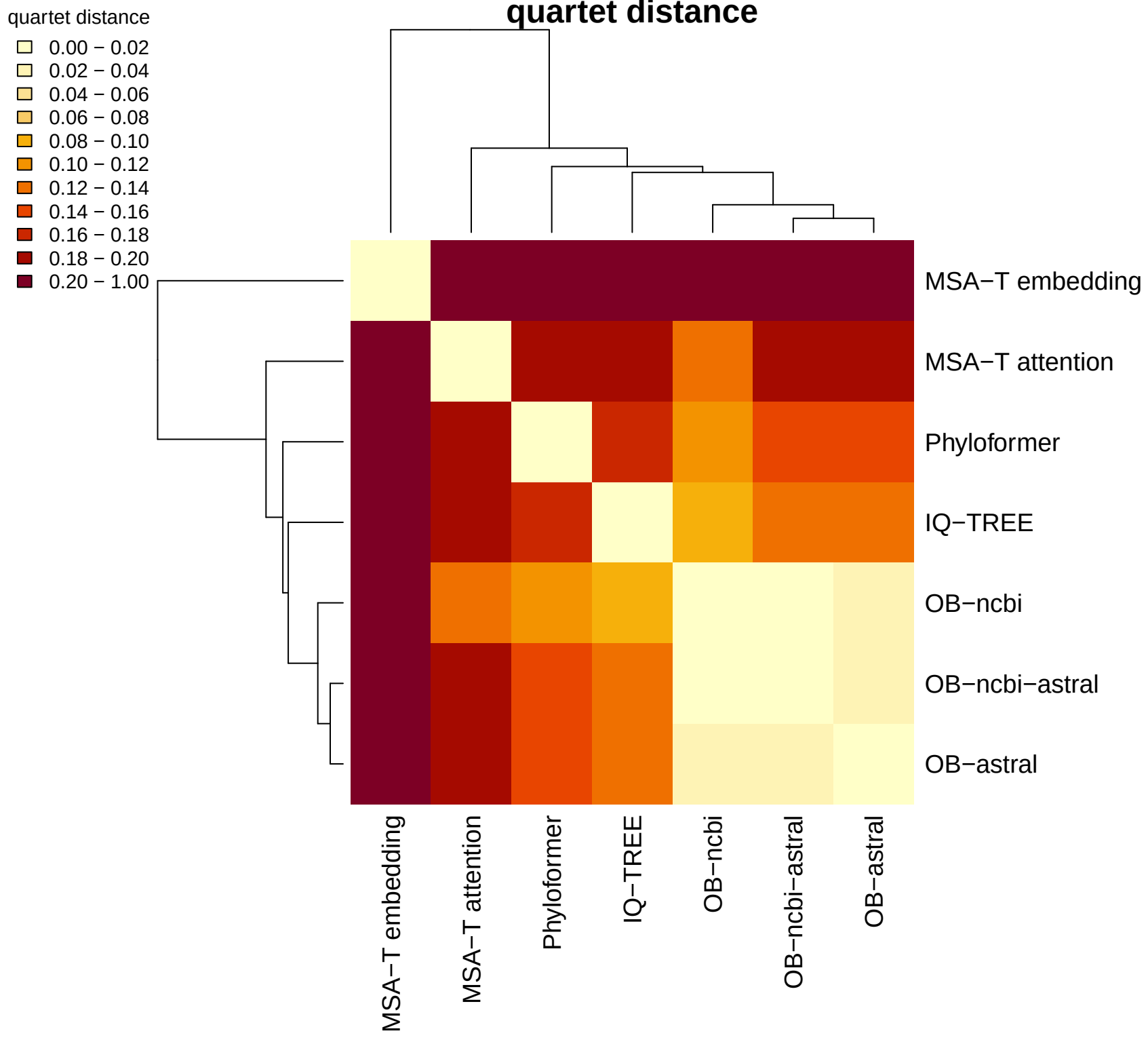


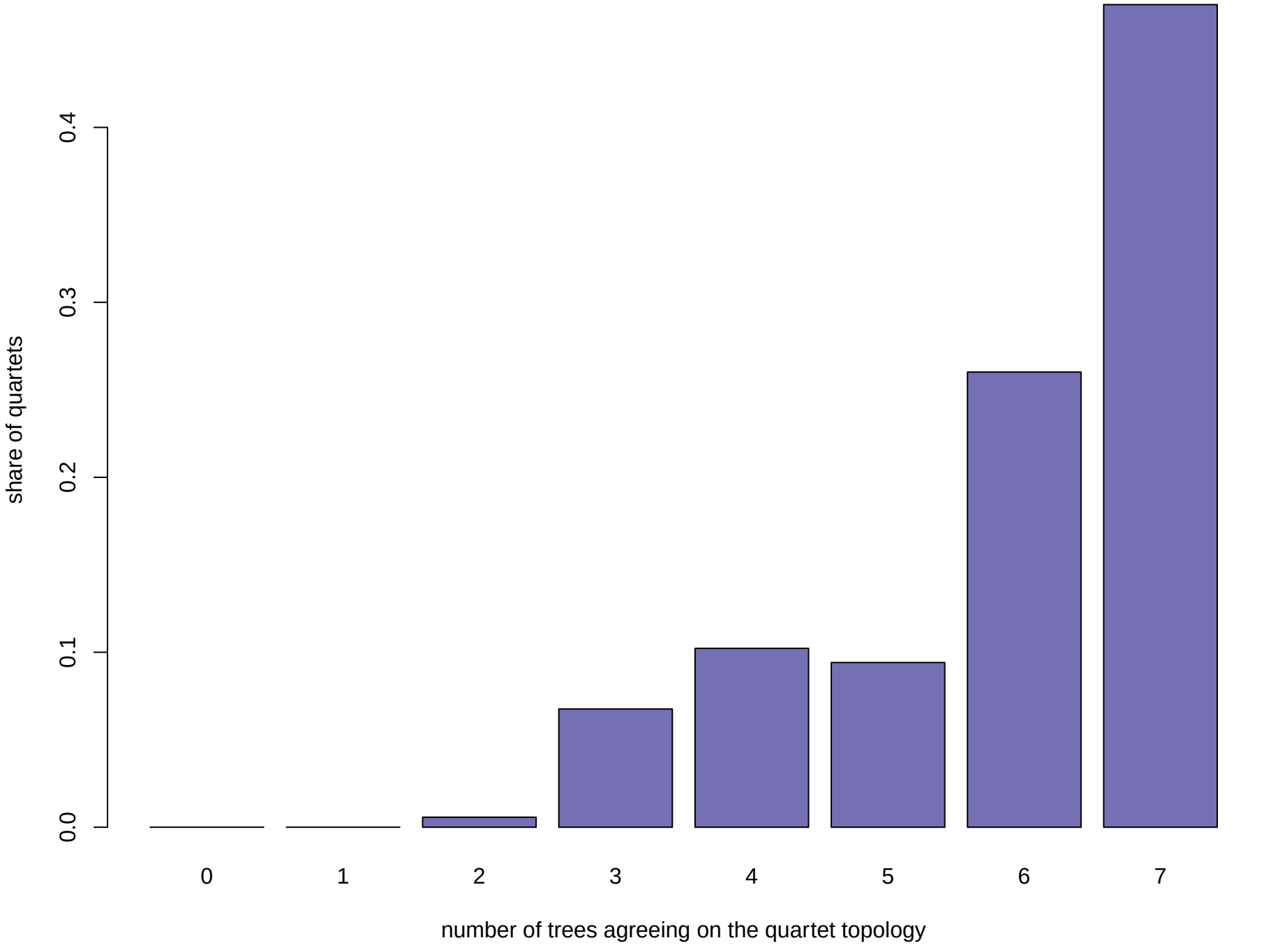
IQ-TREE vs OB-ncbi

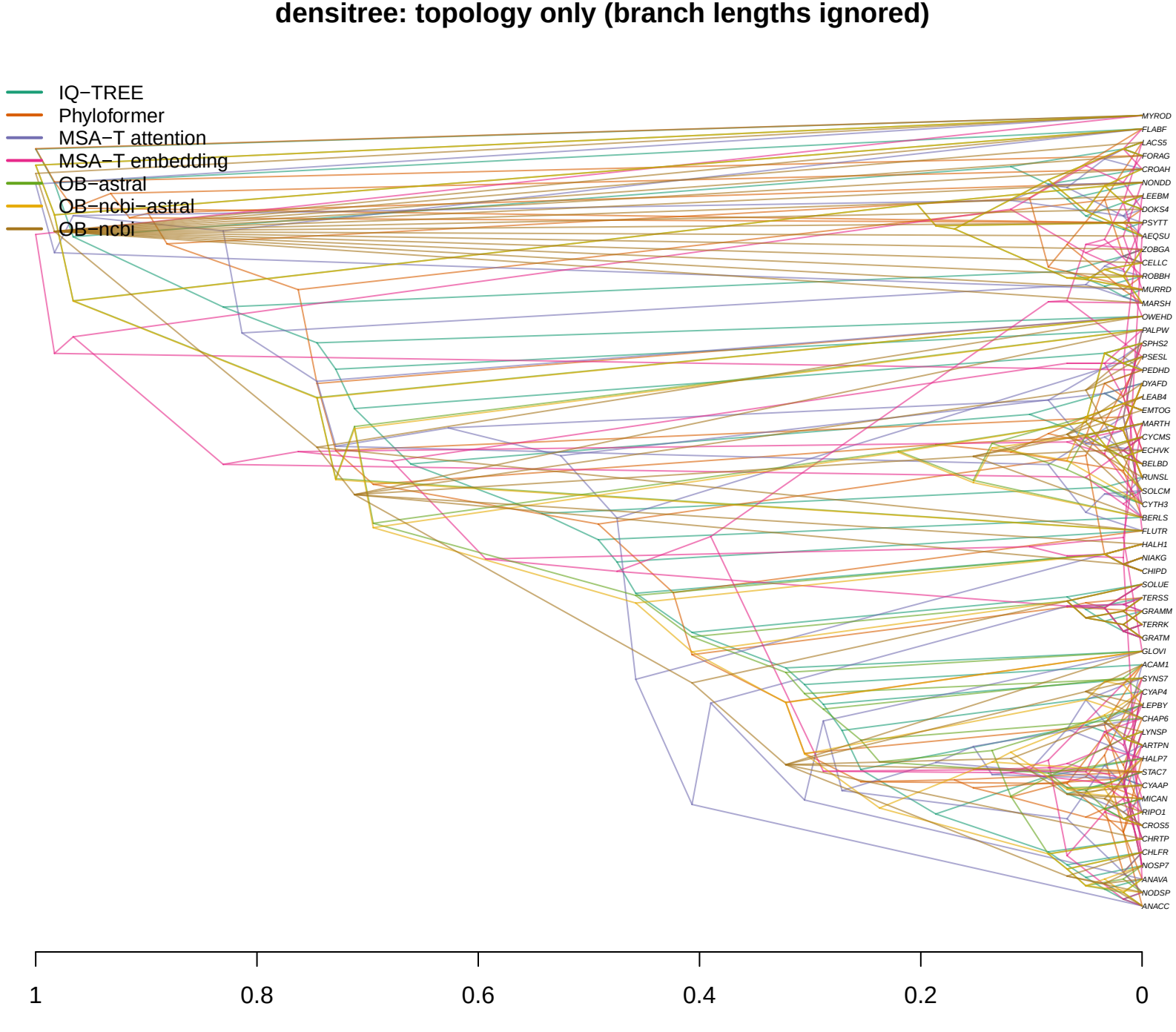


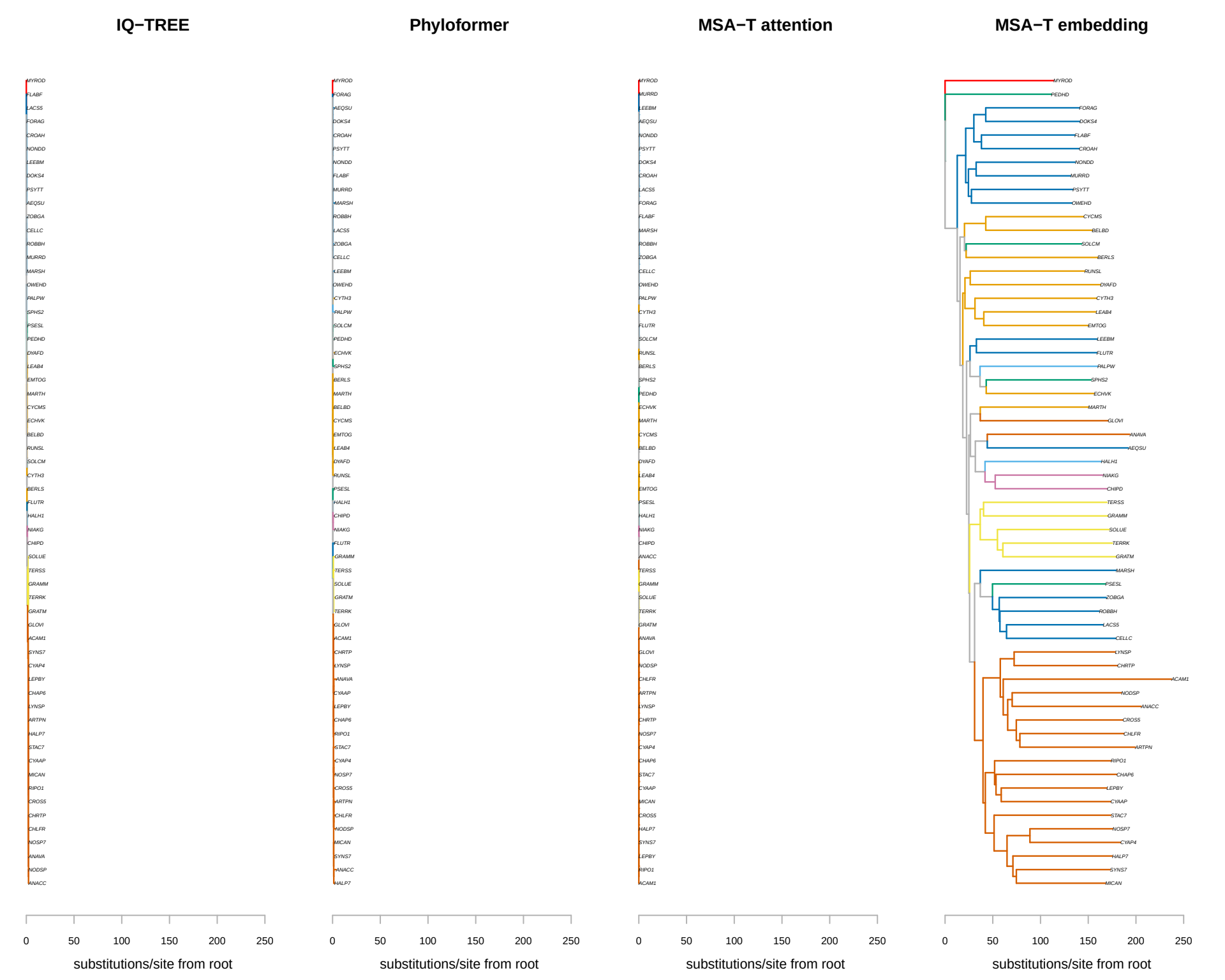

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
MSA−T embedding
Phyloformer
MSA−T attention
OB−ncbi
IQ−TREE
OB−ncbi−astral
OB−astral
MSA−T embedding
Phyloformer
MSA−T attention
OB−ncbi
IQ−TREE
OB−ncbi−astral
OB−astral


methods by RF distance
MSA−T attention
IQ−TREE
Phyloformer
OB−astral
OB−ncbi−astral
OB−ncbi
MSA−T embedding

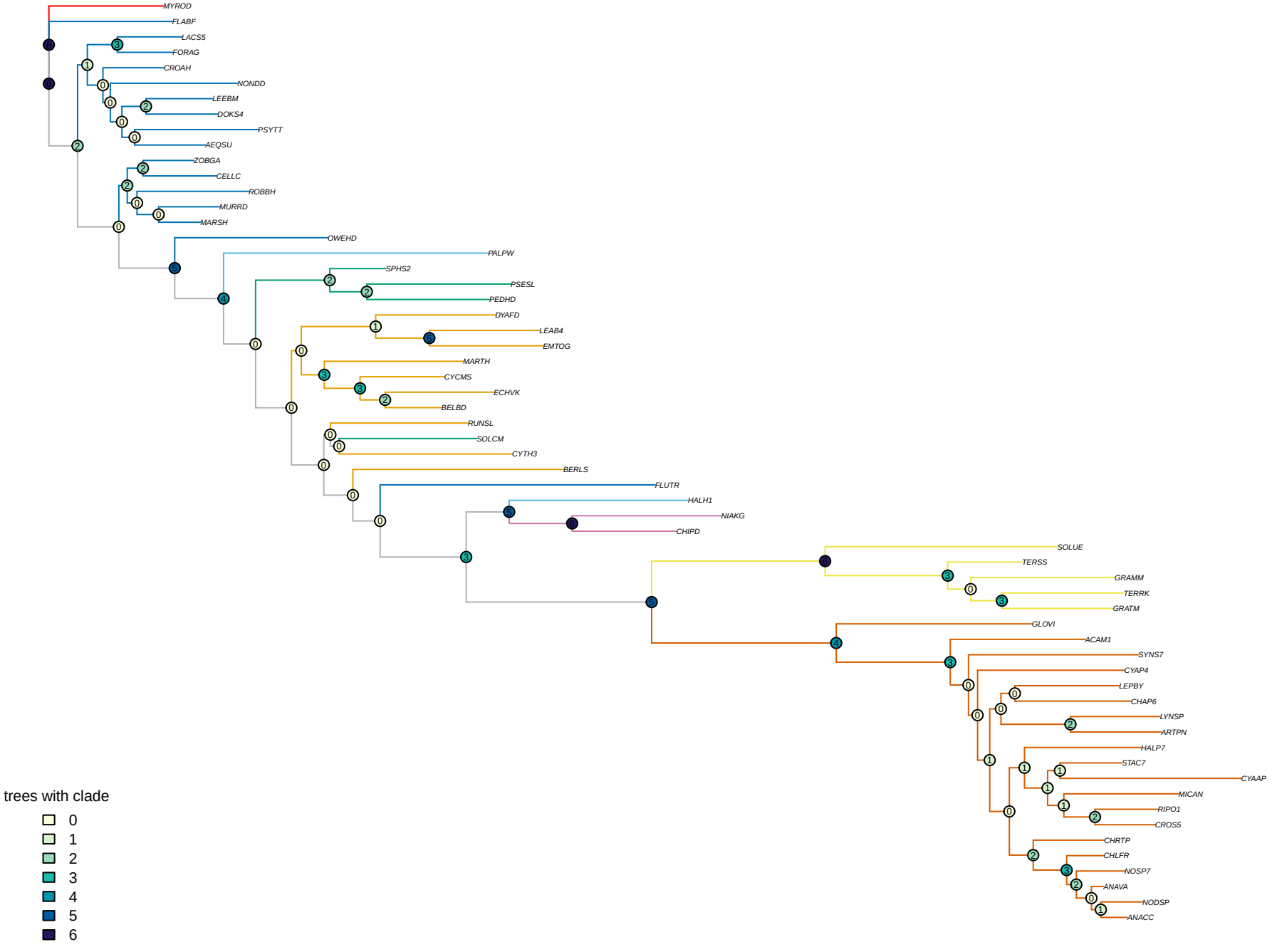

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
MSA−T embedding
MSA−T attention
Phyloformer
IQ−TREE
OB−ncbi
OB−ncbi−astral
OB−astral
MSA−T embedding
MSA−T attention
Phyloformer
IQ−TREE
OB−ncbi
OB−ncbi−astral
OB−astral


quartet consistency (7 trees; 47.0% shared by all)
share of quartets
0.0
0.1
0.2
0.3
0.4
0
1
2
3
4
5
6
7
number of trees agreeing on the quartet topology

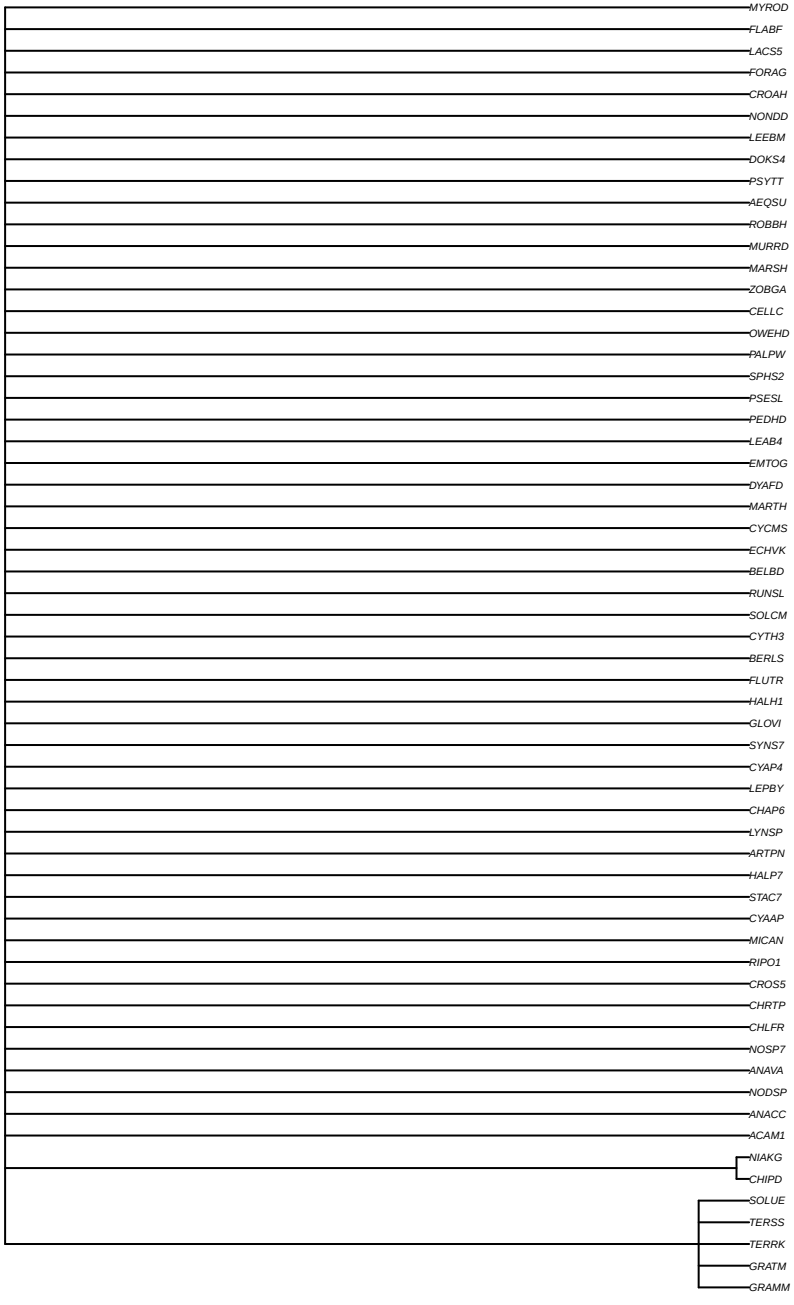

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
MSA−T attention
MSA−T embedding
OB−astral
OB−ncbi−astral
OB−ncbi
MYROD
FLABF
LACS5
FORAG
CROAH
NONDD
LEEBM
DOKS4
PSYTT
AEQSU
ZOBGA
CELLC
ROBBH
MURRD
MARSH
OWEHD
PALPW
SPHS2
PSESL
PEDHD
DYAFD
LEAB4
EMTOG
MARTH
CYCMS
ECHVK
BELBD
RUNSL
SOLCM
CYTH3
BERLS
FLUTR
HALH1
NIAKG
CHIPD
SOLUE
TERSS
GRAMM
TERRK
GRATM
GLOVI
ACAM1
SYNS7
CYAP4
LEPBY
CHAP6
LYNSP
ARTPN
HALP7
STAC7
CYAAP
MICAN
RIPO1
CROS5
CHRTP
CHLFR
NOSP7
ANAVA
NODSP
ANACC
1
0.8
0.6
0.4
0.2
0


IQ−TREE
Phyloformer
MSA−T attention
MSA−T embedding
MYROD
FLABF
LACS5
FORAG
CROAH
NONDD
LEEBM
DOKS4
PSYTT
AEQSU
ZOBGA
CELLC
ROBBH
MURRD
MARSH
OWEHD
PALPW
SPHS2
PSESL
PEDHD
DYAFD
LEAB4
EMTOG
MARTH
CYCMS
ECHVK
BELBD
RUNSL
SOLCM
CYTH3
BERLS
FLUTR
HALH1
NIAKG
CHIPD
SOLUE
TERSS
GRAMM
TERRK
GRATM
GLOVI
ACAM1
SYNS7
CYAP4
LEPBY
CHAP6
LYNSP
ARTPN
HALP7
STAC7
CYAAP
MICAN
RIPO1
CROS5
CHRTP
CHLFR
NOSP7
ANAVA
NODSP
ANACC
0
50
100
150
200
250
substitutions/site from root

IQ−TREE clades: number of the other 6 trees that contain each clade



strict



majority rule

